TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Na_2CO_3 Content in Washing Soda

A standard-acid titration taken all the way to the methyl-orange end point — both of the carbonate's protons counted, and the answer read against the arithmetic of the decahydrate itself.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 12 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the percentage of sodium carbonate in a sample of washing soda. Washing soda is the decahydrate, $Na_2CO_3 \cdot 10H_2O$ (molar mass 286 g mol^{-1}), so even a perfectly pure sample is only 106/286 = 37.1 % Na_2CO_3 by mass — the rest is water of crystallisation. The assay: dissolve a weighed sample, and titrate portions of it with standard hydrochloric acid. Carbonate neutralises acid in **two steps**, and the indicator chooses which step you stop at:

$$\text{Step 1: } CO_3^{2-} + H^+ \rightarrow HCO_3^- \quad \text{(phenolphthalein end point — half-neutralisation)}$$

$$\text{Step 2: } HCO_3^- + H^+ \rightarrow H_2CO_3 \rightarrow H_2O + CO_2 \quad \text{(methyl orange end point — full neutralisation)}$$

$$\text{Overall to methyl orange: } Na_2CO_3 + 2\,HCl \rightarrow 2\,NaCl + H_2O + CO_2 \quad \text{(1 mol carbonate : 2 mol acid)}$$

Because the question asks for the *total* Na_2CO_3 content, the titration is run to the **methyl orange** end point — yellow in the alkaline carbonate solution, turning to the first permanent **orange** when the last hydrogencarbonate is destroyed (the equivalence solution, saturated with carbonic acid, sits near pH 3.7, inside methyl orange's range). The equivalent weight of Na_2CO_3 in this titration is therefore 106/2 = 53.0 g per equivalent — half the molar mass, because both protons are counted. (Had phenolphthalein been used, the equivalent weight would be 106 and the titre exactly halved — same solution, different question.)

2. Apparatus and solutions

- Standard hydrochloric acid, 0.100 M (standardised against anhydrous Na_2CO_3 or borax), in the burette.

- Washing soda sample: 2.86 g dissolved and made up to 250 mL (chosen because pure decahydrate would make the stock exactly 0.0100 mol of Na_2CO_3 units — deviations of the titre from the ideal then speak directly about the sample).
- Methyl orange indicator; pipette 25 mL; conical flask, white tile; a means of boiling the flask briefly (see Procedure).

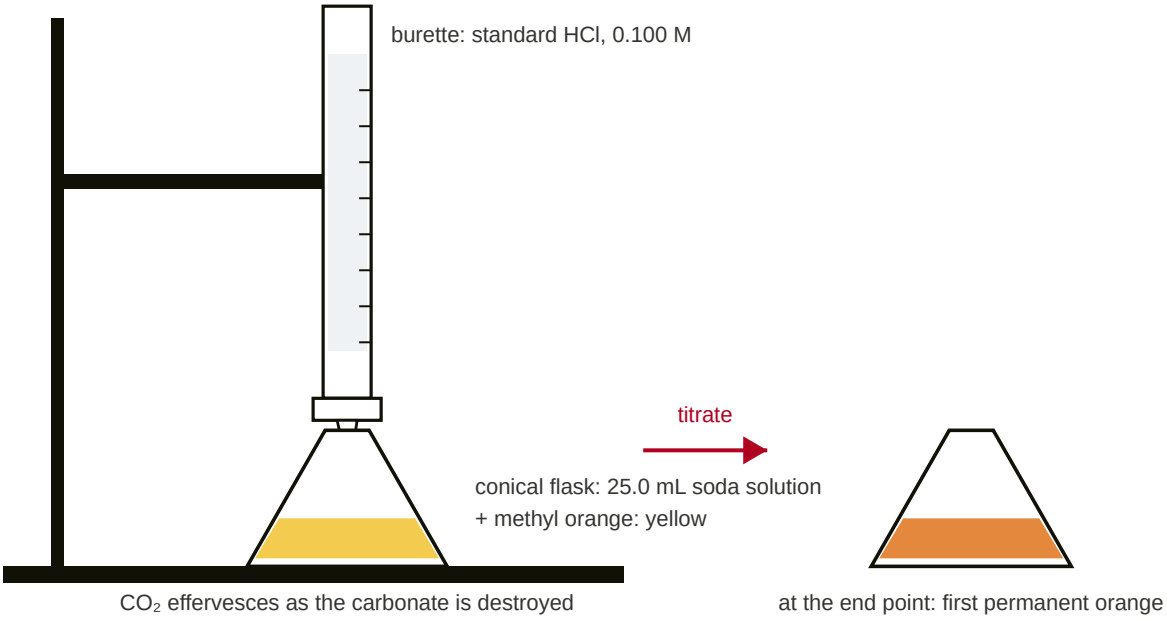


Figure 1 — Titrating washing-soda solution with standard acid to methyl orange. The indicator rides yellow through both neutralisation steps and turns orange only when the carbonate is wholly converted to carbonic acid.

3. Procedure

1. Weigh 2.86 g of the washing soda sample, dissolve it in distilled water, transfer quantitatively to a 250 mL volumetric flask and make up to the mark.
2. Rinse and fill the burette with the standard 0.100 M HCl.
3. Pipette 25.0 mL of the soda solution into a conical flask; add 2–3 drops of methyl orange — the solution is yellow.
4. Titrate with swirling. Carbon dioxide effervesces as the end approaches; slow to drops.
5. **Sharpening trick:** when the colour first begins to turn, boil the flask briefly to expel dissolved CO_2 (which otherwise blurs the change by holding the solution at carbonic-acid pH too early), cool it, and finish the titration dropwise.
6. End point: the first drop that turns the solution **permanently orange** (not red — red is overshoot into strong-acid excess). Record the titre.
7. Repeat until two titres agree within 0.1 mL; average the concordant titres.

4. Worked calculation (model data)

Na_2CO_3 CONTENT RESULT

Sample: 2.86 g washing soda made up to 250 mL. Aliquot: 25.0 mL. Standard HCl: 0.100 M. Concordant titre: 19.4 mL.

Moles of HCl = $0.100 \times 0.0194 = 1.94 \times 10^{-3}$ mol.

Moles of Na_2CO_3 in the aliquot = $1.94 \times 10^{-3} / 2 = 9.70 \times 10^{-4}$ mol (ratio 1 : 2 to methyl orange).

Moles in the whole flask = $9.70 \times 10^{-4} \times 10 = 9.70 \times 10^{-3}$ mol → mass = $9.70 \times 10^{-3} \times 106 = 1.028$ g Na_2CO_3.

Na_2CO_3 content = 1.028/2.86 × 100 = **36.0 %**.

Reading the result: pure $Na_2CO_3 \cdot 10H_2O$ would give 37.1 % (and a titre of 20.0 mL). The sample assays at 36.0 % — close to the theoretical value, that of a good commercial washing soda carrying a little moisture or inert matter. By the equivalents road: 1.94×10^{-3} eq × 53.0 g eq^{-1} = 0.1028 g per aliquot ✓ — the same 1.028 g per flask.

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	19.8	19.8
2	0.0	19.4	19.4
3	0.0	19.4	19.4

The ideal titre for the pure decahydrate at this weighing is 20.0 mL by construction — so the difference between 19.4 and 20.0 mL *is* the purity statement of the sample, visible before any calculating is done.

6. Precautions

- Stop at orange, not red: methyl orange is fully red only in distinctly acid solution, so a red flask is an overshot flask.
- Boil off CO_2 near the end point (and cool before finishing): dissolved carbonic acid drags the indicator toward orange prematurely, giving a creeping, uncertain end point and a low titre.
- Use only 2–3 drops of indicator — methyl orange is itself an acid–base system and a heavy dose shifts the end point.
- Weigh the crystals quickly and keep the sample stoppered: washing soda effloresces in dry air and absorbs moisture in damp air, and either changes what ‘2.86 g of sample’ means between the balance

and the flask.

- The pipette is rinsed with the soda solution, the burette with the acid; the flask with distilled water only.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is methyl orange, and not phenolphthalein, the indicator for the *total* carbonate content?
2. The same 25.0 mL aliquot is titrated with phenolphthalein as indicator. Predict the titre, given the methyl-orange titre of 19.4 mL.
3. A 25.0 mL aliquot (from 2.86 g made to 250 mL) needs 18.2 mL of 0.100 M HCl. Find the Na_2CO_3 percentage.
4. Why does boiling the flask near the end point sharpen the colour change?
5. A washing soda sample is left in an open dish in dry air for a month. When it is assayed, will its Na_2CO_3 percentage come out above or below 37.1 %? Explain.
6. Why is the equivalent weight of Na_2CO_3 53.0 g eq^{-1} in this titration but 106 g eq^{-1} in a phenolphthalein titration?

Answer key

1. Phenolphthalein changes at the first equivalence point, where carbonate has become only hydrogencarbonate — half the acid, half the story. Methyl orange's range (3.1–4.4) brackets the second equivalence point, where neutralisation is complete ($CO_3^{2-} \rightarrow H_2CO_3$), so its titre counts the whole carbonate.
2. Exactly half: 9.7 mL — the phenolphthalein step consumes one proton per carbonate, the methyl-orange journey consumes two.
3. Moles HCl = $0.100 \times 0.0182 = 1.82 \times 10^{-3}$; moles Na_2CO_3 per aliquot = 9.10×10^{-4}; per flask 9.10×10^{-3} mol $\rightarrow$ 0.965 g; content = 0.965/2.86 × 100 = 33.7 %.
4. Dissolved CO_2/carbonic acid is a weak acid that buffers the solution near the end point and starts turning methyl orange before the hydrogencarbonate is truly finished — a gradual, early change. Boiling expels the CO_2, so the last of the HCO_3^- meets the acid unbuffered and the yellow-to-orange turn snaps at the correct drop.
5. Above 37.1 %. Washing soda effloresces: it loses water of crystallisation to dry air, so each gram of the weathered sample contains more Na_2CO_3 and less water than the decahydrate formula assumes — in the limit of fully anhydrous salt it would approach 100 %. (Adsorbed damp in humid air would push it the other way.)

6. Equivalent weight is molar mass divided by the number of protons the formula unit actually accepts *in the titration being done*. To methyl orange each CO_3^{2-} accepts two protons (106/2 = 53.0); to phenolphthalein it accepts only one (106/1 = 106). The equivalent is a property of the reaction, not just of the substance.

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